THE PRODUCT RULE

If $p(x) = f(x)g(x)$, then $p'(x) = f'(x)g(x) + f(x)g'(x)$.

Note: Leibniz Notation: If u and v are functions of x, $\frac{d}{dx}(uv) = \frac{du}{dx}v + u\frac{dv}{dx}$

Proof:

Let $p(x) = f(x)g(x)$, then

$$p'(x) = \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{(x+h)-x}$$

$$= \lim_{h\to 0} \frac{f(x+h)g(x+h) + 0 - f(x)g(x)}{h}$$

Just like multiplying by 1 is a powerful tool in math, so is adding 0
In this case, $0 = -f(x)g(x+h) + f(x)g(x+h) \backslash$

$$= \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h}$$

Factor out $g(x+h)$ from the first two terms of the numerator, and we'll factor out $f(x)$ from the last two terms of the numerator

$$= \lim_{h\to 0} \left\{ \left[\frac{f(x+h) - f(x)}{h} \right] g(x+h) + f(x) \left[\frac{g(x+h) - g(x)}{h} \right] \right\}$$

$$= \lim_{h\to 0} \left[\frac{f(x+h) - f(x)}{h} \right] \lim_{h\to 0} g(x+h) + \lim_{h\to 0} f(x) \lim_{h\to 0} \left[\frac{g(x+h) - g(x)}{h} \right]$$

$$= f'(x)g(x) + f(x)g'(x)$$

Examples

1. Differentiate $h(x) = (x^3 - 2x)(3x^4 + 2x + 8)$ using the product rule

2. Determine the value of $f'(-1)$ for the function $f(x) = (3x^4 - 12x^2 + 4x - 9)(6x^7 - 4x^4 + 18)$

Determine an expression for $p'(x)$ if $p(x) = f(x)g(x)h(x)$.

$$p'(x) =$$

This is called the ***extended product rule*** for three functions.

3. Differentiate the cubic function $f(x) = x(2x+5)(x-1)$ by using the extended product rule.

4. If $g(x) = x^2 f(x)$, $f(2) = -2$ and $g'(2) = 8$, then determine $f'(2)$.

THE QUOTIENT RULE

If $h(x)=\frac{f(x)}{g(x)}$, then $h'(x)=\frac{f'(x)g(x)-f(x)g'(x)}{[g(x)]^2}$, $g(x)\neq 0$.

Note: Leibniz Notation: $\frac{d}{dx}\left(\frac{u}{v}\right)=\frac{\frac{du}{dx}v-u\frac{dv}{dx}}{v^2}$

Proof:

Let $h(x)=\frac{f(x)}{g(x)}$, $g(x)\neq 0$, therefore

$$h(x)g(x)=f(x)$$

$$h'(x)g(x)+h(x)g'(x)=f'(x)$$

$$h'(x)g(x)=f'(x)-h(x)g'(x)$$

$$h'(x)=\frac{f'(x)-h(x)g'(x)}{g(x)}$$

$$=\frac{f'(x)-\frac{f(x)}{g(x)}g'(x)}{g(x)}$$

$$=\frac{f'(x)-\frac{f(x)}{g(x)}g'(x)}{g(x)}\times\frac{g(x)}{g(x)}$$

$$=\frac{f'(x)g(x)-f(x)g'(x)}{[g(x)]^2}$$

Examples

1. Determine the derivative of $h(x)=\dfrac{3x-4}{x^2+5}$.

2. Determine the coordinates of each point on the graph of $f(x)=\dfrac{2x+8}{\sqrt{x}}$ where the tangent is horizontal.

Additional Questions

1. Determine the equation of the tangent line to $g(x)=\left(\frac{1}{x^3}+1\right)(x-1)$ at $x=-1$.

2. If $g(x)=\frac{f(x)}{\sqrt{x-1}}$, where $f(5)=8$, and $f'(5)=-5$, find $g'(5)$.

3. Determine the points on the curve $y=\frac{x}{x+1}$ where the **normal** line is parallel to $x+y=2$.